

Ex. 16. If $x = y^a$, $y = z^b$ and $z = x^c$, then find the value of abc .

Sol. $z^1 = x^c = (y^a)^c \quad [\because x = y^a]$
 $= y^{(ac)} = (z^b)^{ac} \quad [\because y = z^b]$
 $= z^{b(ac)} = z^{abc}$
 $\therefore abc = 1.$

Ex. 17. Simplify : $\left(\frac{x^a}{x^b}\right)^{(a^2+b^2+ab)} \times \left(\frac{x^b}{x^c}\right)^{(b^2+c^2+bc)} \times \left(\frac{x^c}{x^a}\right)^{(c^2+a^2+ca)}.$

Sol. Given Expression $= \{x^{(a-b)}\}^{(a^2+b^2+ab)} \cdot \{x^{(b-c)}\}^{(b^2+c^2+bc)} \cdot \{x^{(c-a)}\}^{(c^2+a^2+ca)}$
 $= x^{(a-b)(a^2+b^2+ab)} \cdot x^{(b-c)(b^2+c^2+bc)} \cdot x^{(c-a)(c^2+a^2+ca)}$
 $= x^{(a^3-b^3)} \cdot x^{(b^3-c^3)} \cdot x^{(c^3-a^3)} = x^{(a^3-b^3+b^3-c^3+c^3-a^3)} = x^0 = 1.$

Ex. 18. If $8^x \cdot 2^y = 512$ and $3^{3x+2y} = 9^6$, then what is the value of x and y ?

(M.A.T., 2004)

Sol. $8^x \cdot 2^y = 512 \Leftrightarrow (2^3)^x \cdot 2^y = 2^9 \Leftrightarrow 2^{3x+y} = 2^9 \Leftrightarrow 3x+y=9 \quad \dots(i)$
 And, $3^{3x+2y} = 9^6 \Leftrightarrow 3^{3x+2y} = (3^2)^6 = 3^{12} \Leftrightarrow 3x+2y=12 \quad \dots(ii)$
 Subtracting (i) from (ii), we get: $y = 3.$
 Putting $y = 3$ in (i), we get: $3x = 6$ or $x = 2.$
 Hence, $x = 2$ and $y = 3.$

Ex. 19. Find the largest from among $\sqrt[4]{6}$, $\sqrt{2}$ and $\sqrt[3]{4}$.

Sol. Given surds are of order 4, 2 and 3 respectively. Their L.C.M. is 12.
 Changing each to a surd of order 12, we get :

$$\sqrt[4]{6} = 6^{\frac{1}{4}} = 6^{\left(\frac{1}{4} \times \frac{3}{3}\right)} = \left(6^{\frac{3}{12}}\right) = (6^3)^{\frac{1}{12}} = (216)^{\frac{1}{12}}.$$

$$\sqrt{2} = 2^{\frac{1}{2}} = 2^{\left(\frac{1}{2} \times \frac{6}{6}\right)} = \left(2^{\frac{6}{12}}\right) = (2^6)^{\frac{1}{12}} = (64)^{\frac{1}{12}}.$$

$$\sqrt[3]{4} = 4^{\frac{1}{3}} = 4^{\left(\frac{1}{3} \times \frac{4}{4}\right)} = \left(4^{\frac{4}{12}}\right) = (4^4)^{\frac{1}{12}} = (256)^{\frac{1}{12}}.$$

$$\text{Clearly, } (256)^{\frac{1}{12}} > (216)^{\frac{1}{12}} > (64)^{\frac{1}{12}}.$$

$$\therefore \text{Largest one is } (256)^{\frac{1}{12}} \text{ i.e., } \sqrt[3]{4}.$$

Ex. 20. Find the square root of $(3 + \sqrt{5})$.

(I.I.C.A.A.O., 2007)

Sol. $\sqrt{3+\sqrt{5}} = \sqrt{3+2\sqrt{\frac{5}{4}}} = \sqrt{\frac{5}{2} + \frac{1}{2} + 2\sqrt{\frac{5}{2} \times \frac{1}{2}}} = \sqrt{\left(\sqrt{\frac{5}{2}}\right)^2 + \left(\sqrt{\frac{1}{2}}\right)^2 + 2\sqrt{\frac{5}{2}}\sqrt{\frac{1}{2}}}$
 $= \sqrt{\left(\sqrt{\frac{5}{2}} + \sqrt{\frac{1}{2}}\right)^2} = \left(\sqrt{\frac{5}{2}} + \sqrt{\frac{1}{2}}\right).$

Ex. 21. If $x = 3 + 2\sqrt{2}$, find the value of $\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right).$

Sol. $\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 = x + \frac{1}{x} - 2 = (3 + 2\sqrt{2}) + \frac{1}{(3 + 2\sqrt{2})} - 2$
 $= (3 + 2\sqrt{2}) + \frac{1}{(3 + 2\sqrt{2})} \times \frac{(3 - 2\sqrt{2})}{(3 - 2\sqrt{2})} - 2 = (3 + 2\sqrt{2}) + (3 - 2\sqrt{2}) - 2 = 4.$
 $\therefore \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right) = 2.$